

Yiming Liu, Ph.D.

Rice Academy Postdoctoral Fellow

Department of Civil and Environmental Engineering, Rice University

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Summary

My research addresses grand challenges in **water**, (critical) **minerals**, and **energy circularity**, with a particular focus on recovering valuable resources from industrial and natural brines. I have developed an integrated understanding of **interfacial science**, **electrochemistry**, and **membrane science**, together with skillsets in separation material synthesis, electrochemical engineering, membrane process engineering, physics-based simulations, and water chemistry. I have applied these capabilities to answer key research questions within the areas of critical mineral recovery, prevention of inorganic precipitation, and waste brine minimization.

Education

- 2023 **University of California, Los Angeles (UCLA)**
 Ph.D. in Civil and Environmental Engineering
 Dissertation: Exploiting Interfacial Energy Transfer for Sustainable Brine Management
 Advisor: Prof. David Jassby, Co-Advisor: Prof. Eric M.V. Hoek
- 2019 **University of California, Los Angeles (UCLA)**
 M.S. in Civil and Environmental Engineering
 Selected Chemical and Biomolecular Engineering Coursework (all A/A+): CH ENGR 101C *Mass Transfer*; CH ENGR 106 *Chemical Reaction Engineering*; CH ENGR C212 *Polymer Processes*; CH ENGR C216 *Surface & Interface Engineering*
- 2018 **Tsinghua University**
 B.E. in Environmental Engineering (ABET-accredited)
 Academic Excellence Award Winners (Twice)
 Senior Thesis Advisor: Prof. Gang Yu, CAE (Elected Member in Environmental Engineering Division, Chinese Academy of Engineering (CAE))

Research Experience

Rice Academy Postdoctoral Fellow	Rice University
Advisor: Prof. Qilin Li, Mentor: Prof. Pedro J.J. Alvarez, NAE	2023.8-present
<i>Selective electrodialysis for extraction of lithium from salt lake brine</i>	2025.9-present
Assessed next-generation ion-exchange membranes and identified performance limits governing selective lithium recovery from complex brines.	
<i>Metal-organic frameworks for recovery of rare earth elements (REEs)</i>	2024.2-present
Designed and synthesized platform (metal-organic frameworks) MOFs enabling high-density organophosphate ligand functionalization for enhanced REE binding.	

- Demonstrated efficient REE extraction from challenging waste matrices and quantified separation performance across competing ions.
- Led molecular dynamics simulation collaboration to reveal the ligand–REE coordination mechanisms controlling selectivity under variable loadings and water chemistries.

Electrochemical mineral scale prevention 2023.8-2024.1

- Developed and validated mechanistic models showing how AC-induced electrokinetic forces suppression of nucleation and growth of calcite crystals.

Graduate Student Researcher

UCLA

Advisors: Profs. David Jassby & Eric M.V. Hoek

2019.6-2023.6

Electrochemical mineral scale prevention

2022.7-2023.6

- Invented an alternating-currents(AC)-driven antiscaling strategy that substantially reduces calcite formation on electrically active surfaces and its precipitation in the bulk.

Direct-heated membranes for thermal brine concentration

2019.6-2022.7

- Created a membrane-based thermal concentration process capable of treating hypersaline brines and produced water with enhanced energy efficiency.
- Built multiscale simulations (module-level heat/mass transfer and system-level energy analysis) to optimize operational design for water recovery and energy consumption.

Honors & Awards

2026	Founding President Best Paper Award, Chinese-American Professors in Environmental Engineering and Science (CAPEES)
2025	Outstanding Oral Presentation Award, 19th Chinese Environmental Conference
2023	Rice Academy Postdoctoral Fellowship, Rice University
2023	University Fellowship, UCLA
2021	Internet of Water Idea Challenge Second Place Winner, Industrie De Nora S.p.A.
2019	Cathay Bank Scholarship (Southern California, United States)
2018	Academic Excellence Award (Twice), Tsinghua University
2017	Civil Centennial Future Scholars, The University of Hong Kong

Proposal Writing Experience

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- Advanced Research Projects Agency – Energy (ARPA-E), U.S. Department of Energy, *Selective extraction of rare earth elements from mining wastewater using electrodialysis enabled by novel thin film nanocomposite cation exchange membranes*, PI: Qilin Li, Rice University. **Role: Drafting 90% of the proposal; editing, revising, and proof-reading the draft. Prepared key figures.** December 2024.
 - Boomerang’s carbon capture research grant, Baydin, Inc., *Simultaneous carbon capture and direct lithium extraction*, PIs: Yiming Liu and Yuren Feng, Rice University. **Role: Drafting 70% of the proposal and submitting the final version.** December 2023.
 - Rice Academy Postdoctoral Fellowship, Rice University Office of Graduate and Postdoctoral Studies, *Selective ion separation and green hydrogen generation technologies to enhance energy transition*, PIs: Yiming Liu and Qilin Li (Funded). **Role: Drafting the proposal and submitting the final version.** December 2022.

Publications

Published (¹: Co-first author, *: Corresponding author)

1. Zhu, Y.^¹, Liu, Y.^¹, Weng, X., Lee, J., Feng, Y., **Liu, Y.**, Xie, T., Tu, W., Li, Q., Lou, J.* Structure engineering of covalent organic frameworks as metal-free sorbents for urea recovery. *Journal of Materials Chemistry A*, 2025. ([link](#))
2. Feng, Y.^¹, Zhu, Y.^¹, Chen, W., Huang, X., Weng, X., Meyer, M.D., Chen, T., **Liu, Y.**, He, Z., Hou, C., Zuo, K., Yip, N.Y., Gong, K., Lou, J.*^¹, Li, Q.* A rationally designed scalable thin film nanocomposite cation exchange membrane for precise lithium extraction. *Nature Communications*, 16, 1, 8618, 2025 ([link](#))
3. Zhu, Y.^¹, Chen, M.T.^¹, Shen, H., Feng, Y., Ai, Q., **Liu, Y.**, Yan, Y., Li, Q.*^¹, Lou, J.* Hierarchical porous PbO₂ electrode for electro-degradation of various contaminants. *Small Structures*, 2400389, 2025. ([link](#))
4. Dickinson-Cove, A., Plante, E.L., **Liu, Y.**, Hoek, E.M.V., Simonetti, D., Sant, G.*^¹, Jassby, D.* Reactive carbon capture using saline water: evaluation of prospective sources, processes, and products. *Chemical Society Reviews*, 54, 116-151, 2025. ([link](#))
5. **Liu, Y.**^¹, Xiao, M.^¹, Huang, X., Park, J., Hoffman, M.E., Feng, Y., An, A.K., Li, Q., Hoek, E.M.V.*^¹, Jassby, D.* Mitigating CaCO₃ crystal nucleation and growth through continuous ion displacement via alternating electric fields. *Nature Communications*, 16, 35, 2025. ([link](#))

Covered by International Water Association (IWA) in Online Webinar On-Demand:
<https://www.iwa-network.org/learn/advancing-sustainability-with-materials-membranes-electrochemistry/>

Selected as Outstanding Oral Presentation Award in 19th National Environmental Conference (China)

Selected as Founding President Best Paper Award by Chinese-American Professors in Environmental Engineering and Science (CAPEES)

6. Leão, A., Collin, M., Ghodkhande, S., Bouissonnie, A., Chen, X., Malin, B., **Liu, Y.**, Hovey, G., Govindhakannan, J., Plante, E.L., Rosner, F., Jassby, D., Gädt, T., Corsini, L., Simonetti, D., Sant, G.* ZeroCAL: Eliminating carbon dioxide emissions from limestone's decomposition to decarbonize cement production. *ACS Sustainable Chemistry & Engineering*, 12, 43, 15762–15787, 2024. ([link](#))
7. Municchi, F., **Liu, Y.**, Wang, J., Cath, T.Y., Turchi, C.S., Heeley, M.B., Hoek, E.M.V., Jassby, D., Tilton, N.* A numerical method for simulating variable density flows in membrane desalination systems. *Computers & Fluids*, 285, 106449, 2024. ([link](#))
8. Lu, S., McGaughey, A., Im, S., **Liu, Y.**, Wang, X., Leininger, A., Jassby, D., Hoek, E., Ren, Z.J.* Membrane electrolysis distillation for volatile fatty acids extraction from pH-neutral fermented wastewater. *Water Research*, 265, 122306, 2024. ([link](#))
9. Mabry, M., Municchi, F., **Liu, Y.**, Wang, J., Cath, T.Y., Turchi, C.S., Heeley, M.B., Hoek, E.M.V., Jassby, D., Martinand, D., Tilton, N.* Generating buoyancy-driven convection in membrane distillation. *Journal of Membrane Science*, 709, 123043, 2024. ([link](#))
10. Heeley, M.B., **Liu, Y.**, Wang, J., Municchi, F., Lou, J., Dudley, M., Kaddoura, M., Hoek, E.M.V., Tilton, N., Turchi, C.S., Cath, T.Y., Jassby, D.* Economic assessment of a

- multistage surface-heated vacuum membrane distillation process for the treatment of hypersaline produced water. *Desalination*, 582, 117648, 2024. ([link](#))
11. Lou, J., Dudley, M., Wang, J., **Liu, Y.**, Cath, T.Y., Turchi, C.S., Heeley, M.B., Hoek, E.M.V., Jassby, D., Tilton, N.* Numerical simulations of membrane distillation systems with actively heated membranes. *Journal of Membrane Science*, 668, 121206, 2023. ([link](#))
 12. **Liu, Y.**, Wang, J., Hoek, E.M.V., Municchi, F., Tilton, N., Cath, T.Y., Turchi, C.S., Heeley, M.B., Jassby, D.* Multistage surface-heated vacuum membrane distillation process enables high water recovery and excellent heat utilization: A modeling study. *Environmental Science & Technology*, 57, 1, 643-654, 2023. ([link](#))
 13. **Liu, Y.**, Wang, J., Jung, B., Rao, U., Sedighi, E., Hoek, E.M.V., Tilton, N., Cath, T.Y., Turchi, C.S., Heeley, M.B., Ju, Y.S., Jassby, D.* Desalinating a real hyper-saline pre-treated produced water via direct-heat vacuum membrane distillation. *Water Research*, 218, 118503, 2022. ([link](#))
 14. Wu, J., Wang, J., **Liu, Y.**, Rao, U., Sustainable desalination and water reuse, *Synthesis Lectures on Sustainable Development 2*, (2), 1-204, 2021. Editors: Hoek, E.M.V., Jassby, D., Kaner, R.B., Morgan & Claypool Publishers, San Rafael, CA. ([link](#))
 15. Wang, J.¹, **Liu, Y.**¹(Co-first author), Rao, U., Dudley, M., Ebrahimi, N.D., Lou, J., Han, F., Hoek, E.M.V., Tilton, N., Cath, T.Y., Turchi, C.S., Heeley, M.B., Ju, Y.S., Jassby, D.* Conducting thermal energy to the membrane/water interface for the enhanced desalination of hypersaline brines using membrane distillation. *Journal of Membrane Science*, 626, 119188, 2021. ([link](#))
 16. Ren, C., Yang, P., Gao, J., Huo, X., Min, X., Bi, E.Y., **Liu, Y.**, Wang, Y., Zhu, M., Liu, J.* Catalytic reduction of aqueous chlorate with MoO_x immobilized on Pd/C. *ACS Catalysis*, 10, 15, 8201-8211, 2020. ([link](#))
 17. Feng, Y., Liao, C., Kong, L., Wu, D., **Liu, Y.**, Lee, P., Shih, K.* Facile synthesis of highly reactive and stable Fe-doped g-C₃N₄ composites for peroxymonosulfate activation: a novel nonradical oxidation process. *Journal of Hazardous Materials*, 354, 63-71, 2018. ([link](#))

Preprints

1. **Liu, Y.**, Tang, Y., Chen, W., Zhu, Y., Li, W., Lou, J., Gong, K., Alvarez, P. J. J., Li, Q.* Aqueous adsorption of rare earth elements by organophosphonate-ligand-modified metal-organic frameworks: Role of ligand loading in surface complexation. *Under review, ACS Applied Materials & Interfaces*, 2026 (manuscript available upon request).
2. Zhu, Y.¹, Tian, X.¹, Huang, X.¹, Chen, M.T., Ai, Q., Feng, Y., Yan, Y., Liu, Y., Shin, B., **Liu, Y.**, Xie, T., Tu, W., Zhai, T., Fang, Q., Steinbach, D., Han, Y., Li, Q.*, Lou, J.* Superior anti-scaling performances of pH responsive polymer coated carbon nanotubes. *Under review* (manuscript available upon request).

Manuscripts in Preparation

3. **Liu, Y.**, Chen, W., Tang, Y., Zhu, Y., Gong, K., Lou, J., Li, Q.* Selective recovery of rare earth elements from waste streams by ligand-functionalized metal-organic frameworks: mechanistic and molecular insights into high selectivity.
4. **Liu, Y.**, Tang, Y., Li, Q.* Selective electrodialysis for lithium extraction from brine: Role of operating conditions and water chemistry.

5. **Liu, Y.,** Zhu, Y., Lou, J., Li, Q.* Metal-organic frameworks for ligand-based recovery of rare earth elements (Review).

Presentations

Invited Talks

1. Department of Civil and Environmental Engineering, South Dakota State University, **April 2026**, *Exploiting the Interactions at Solid-Water Interfaces for the Water & Critical Mineral Circularity.*
2. Department of Civil and Environmental Engineering, Lehigh University, **April 2026**, *Exploiting the Interactions at Solid-Water Interfaces for the Water, Energy, & Mineral Circularity.*
3. School of Chemical and Biomolecular Engineering, The University of Sydney, Australia, **March 2026**, *Exploiting the Interactions at Solid-Water Interfaces for the Water, Energy, & Mineral Circularity.*
4. Department of Civil and Environmental Engineering, Vanderbilt University, **February 2026**, *Exploiting the Interactions at Solid-Water Interfaces for the Water, Energy, & Mineral Circularity.*
5. School of Environment, Tsinghua University, **December 2025**, *Exploiting the material interfaces of electrochemistry & membranes for mineral, water & energy circularity.*
6. International Water Association (IWA) Nano & Water Specialist Group, **May 2025**, *Advancing sustainability with materials, membranes & electrochemistry.*
7. College of Environmental Sciences and Engineering, Peking University, **March 2025**, *Interfacial processes for resource extraction and sustainable management of brines.*
8. CEE Departmental Seminar: Lightning Rounds, Rice University, **March 2024**, *Mitigation of mineral crystal nucleation through continuous ion displacement using alternating electric fields.*
9. Rice Academy Interview Symposium, Rice University, **February 2023**, *Selective ion separation and green hydrogen generation technologies to enhance energy transition.*

Oral Conference Presentations

10. 14th Sustainable Nanotechnology Organization Conference, Session 6A Future of Metal-Organic Framework (MOF) in the Nano World: Celebrating 2025 Nobel Prize, **November 2025**, *Metal-organic frameworks for ligand-based recovery of rare earth elements.*
11. 14th Sustainable Nanotechnology Organization Conference, Session 1A Tribute to NEWT, **November 2025**, *Adsorption of magnet-relevant rare earth elements by organophosphonate-ligand-modified metal-organic frameworks: the role of ligand loading*
12. AEESP 2025 Research and Education Conference, Session 19 Resource Recovery from Waste Streams Towards Circular Economy, **May 2025**, *Selective extraction of rare earth elements using organophosphorus ligand modified metal organic frameworks.*
13. 19th National Environmental Conference for Doctoral Students and Postdoctoral Researchers (China), **May 2025**, *Mitigating CaCO₃ nucleation and growth through continuous ion displacement via alternating electric fields.*

14. 2024 Fall ACS meeting, Division of Environmental Chemistry, **August 2024**, *Mitigation of mineral crystal nucleation through continuous ion displacement using alternating electric fields.*
15. 2021 The North America Membrane Society, **September 2021**, *Evaluation of direct heated vacuum membrane distillation process using module-scale simulation.*
16. 2020 Fall ACS meeting, Division of Environmental Chemistry, **August 2020**, *Conducting thermal energy to the membrane/water interface for the enhanced desalination of hypersaline brines using membrane distillation.*

Poster Presentations

17. Rice Sustainability Institute-ExxonMobil Workshop, **September 2025**, *Selective extraction of rare earth elements using organophosphorus ligand modified metal organic frameworks.*
18. 2024 US Army Engineer Research and Development Center- Rice University National Security Research Accelerator Meeting, **August 2024**, *Prevention of mineral crystal nucleation by alternating currents.*
19. 2023 NSF Nano-Enabled Water Treatment (NEWT) Center Mid-Year Meeting, **December 2023**, *Prevention of mineral crystal nucleation by alternating currents.*
20. School of Engineering Research Fair, UCLA, **February 2023**, *Desalinating a real hyper-saline pre-treated produced water via direct-heat vacuum membrane distillation*
21. 2022 Gordon Research Conference: Membrane Materials and Processes, **August 2022**, *Desalinating a real hyper-saline pre-treated produced water via direct-heat vacuum membrane distillation.*
22. 2021 The North America Membrane Society, **September 2021**, *Desalinating a real hyper-saline pre-treated produced water via direct-heat vacuum membrane distillation.*

Teaching & Mentoring Experience

Guest Lecturer

Rice University

- 2025 Fall CEVE 434/534 Fate and Transport of Contaminants in the Environment (Class Size: 12 students; Instructor: Prof. Qilin Li)

UCLA

- 2023 Winter C&EE 255A Physical and Chemical Processes for Water and Wastewater Treatment (Class Size: 21 students; Instructor: Prof. David Jassby)

Research Mentor

Rice University

- Yingxin Tang (PhD student in Civil and Environmental Engineering, Rice University, 2024 fall-now)
- Connor Hanula (Undergraduate student in Chemical and Biomolecular Engineering, Rice University, Summer Undergraduate Research Fellowship, 2025 summer-2026 spring)
- Aaron Wu (High school student, St. John's School, Houston, 2025 summer-2026 spring)
- Nimah Shukkoor (Undergraduate student in Chemistry, Rice University, 2024 spring)

- Pani Raja Abadi (Undergraduate student in Biology, Lone Star College, NSF Research Experiences for Undergraduates Program, 2024 summer, now undergraduate at University of Texas at Austin)

UCLA

- Matthew E. Hoffman (Master's Student in Civil and Environmental Engineering, UCLA 2023 spring, now Civil Engineering Assistant in City of Santa Monica)
- Jane Park (Undergraduate student in Chemical and Biomolecular Engineering, UCLA, 2023 spring, summer)

Service & Leadership

- 3rd Chinese-American Professors in Environmental Engineering and Science Annual Conference, (Expected) May 2026: **Organizing committee member**
- Renewable Natural Resources Foundation, December 2025: **Grant reviewer**
- 14th Sustainable Nanotechnology Organization Conference, November 2025: **Co-chair for Session 1A Tribute to NEWT**
- Gates Foundation Workshop on Distributed Water Reuse and Ideation for the proposed Rice Water Institute Excellence Center of Household Water and Wastewater Management, November 2025 (Rice University & Arizona State University): **Co-organizer**
- AEESP 2025 Research and Education Conference, May 2025: **Poster judge**
- International Water Association Nano & Water Specialist Group, March 2025 (-present): **Early-career professionals' chair**, promoting the specialist group among young scientists
- Consulting with SolMem (Clean Technology Startup) Consulting, February 2025, December 2024, October 2024: **Technical adviser** on water quality analysis

Professional Membership

American Chemical Society, Association of Environmental Engineering and Science Professors, International Water Association, Association of Chinese-American Professors in Environmental Engineering and Science, North American Membrane Society

Peer-Review Experience

Reviewer

Environmental Science & Technology (1), *Desalination* (5), *Separation & Purification Technology* (1), *Journal of Water Process Engineering* (1), *Colloids and Surfaces C: Environmental Aspects* (1), *Desalination and Water Treatment* (3), *Membranes* (2), *Water* (2)

Assisted Reviewer

Journal of Membrane Science (1), *Industrial & Engineering Chemistry Research* (1)

References

1. Dr. Qilin Li (Postdoctoral advisor)

Karl F. Hasselmann Professor, Department of Civil and Environmental Engineering, Department of Chemical and Biomolecular Engineering, and Department of Materials Science and NanoEngineering, Rice University

Thrust Leader, Rice WaTER Institute

Co-Director, Engineering Research Center on Nanotechnology-Enabled Water Treatment (NEWT) Center, NSF

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2. Dr. Pedro J.J. Alvarez, NAE (Postdoctoral mentor and collaborator)

George R. Brown Professor, Department of Civil and Environmental Engineering, Department of Chemical and Biomolecular Engineering, Department of Materials Science and NanoEngineering, and Baker Institute for Public Policy, Rice University

Member, National Academy of Engineering (United States), Chinese Academy of Engineering, and American Academy of Arts and Sciences

Director, Rice WaTER Institute

Founding Director, NEWT Center, NSF

Web: <https://alvarez.rice.edu>, E-mail: alvarez@rice.edu, Tel: (713)-348-5903

3. Dr. David Jassby (Ph.D. advisor)

Professor, Department of Civil and Environmental Engineering, UCLA

Associate Director, Institute for Carbon Management, UCLA

Web: <https://samueli.ucla.edu/people/david-jassby/>, E-mail: jassby@ucla.edu, Tel: (310)-825-1346

4. Dr. Eric M.V. Hoek (Ph.D. co-advisor)

Professor, Department of Civil and Environmental Engineering, UCLA

Faculty Scientist, Energy Storage & Distributed Resources Division, Lawrence Berkeley National Lab

Web: <https://samueli.ucla.edu/people/eric-hoek/>, E-mail: emvhoek@ucla.edu, Tel: (310)-794-7124

5. Dr. Kai Gong (Postdoctoral research collaborator)

Assistant Professor, Department of Civil and Environmental Engineering

Member, Rice Advanced Materials Institute

Member, Ken Kennedy Institute for AI and Computing, Rice University

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